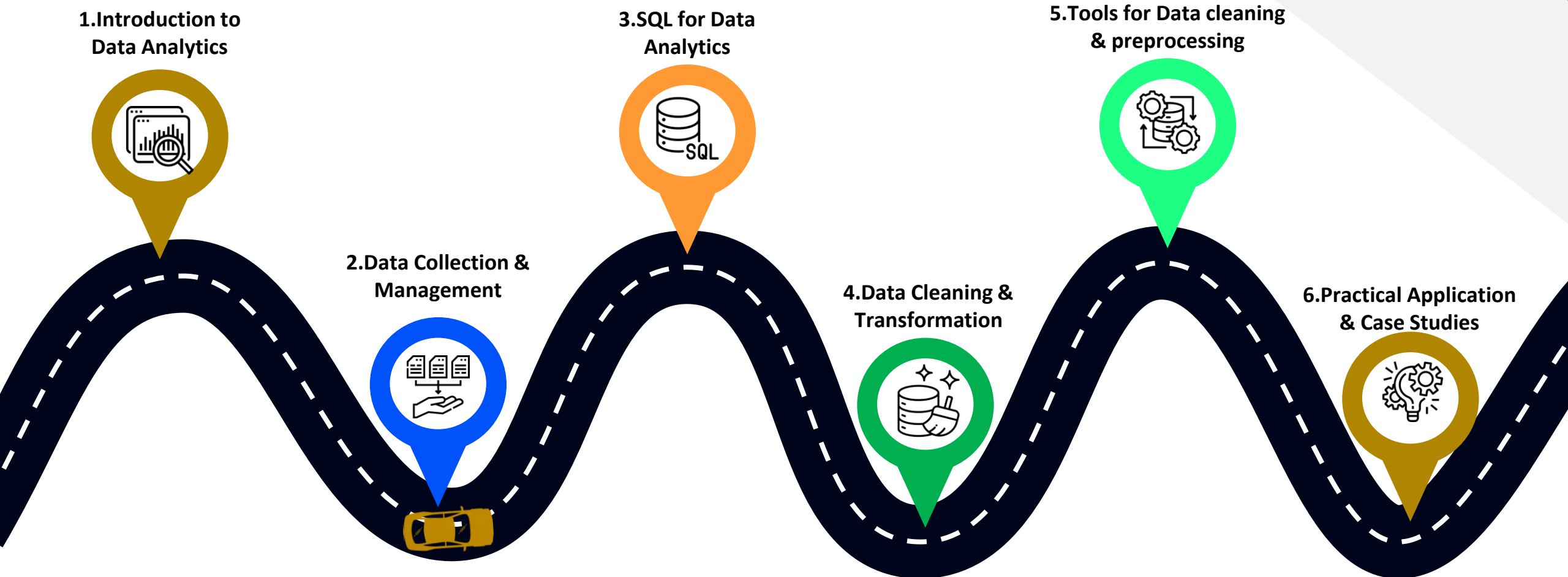




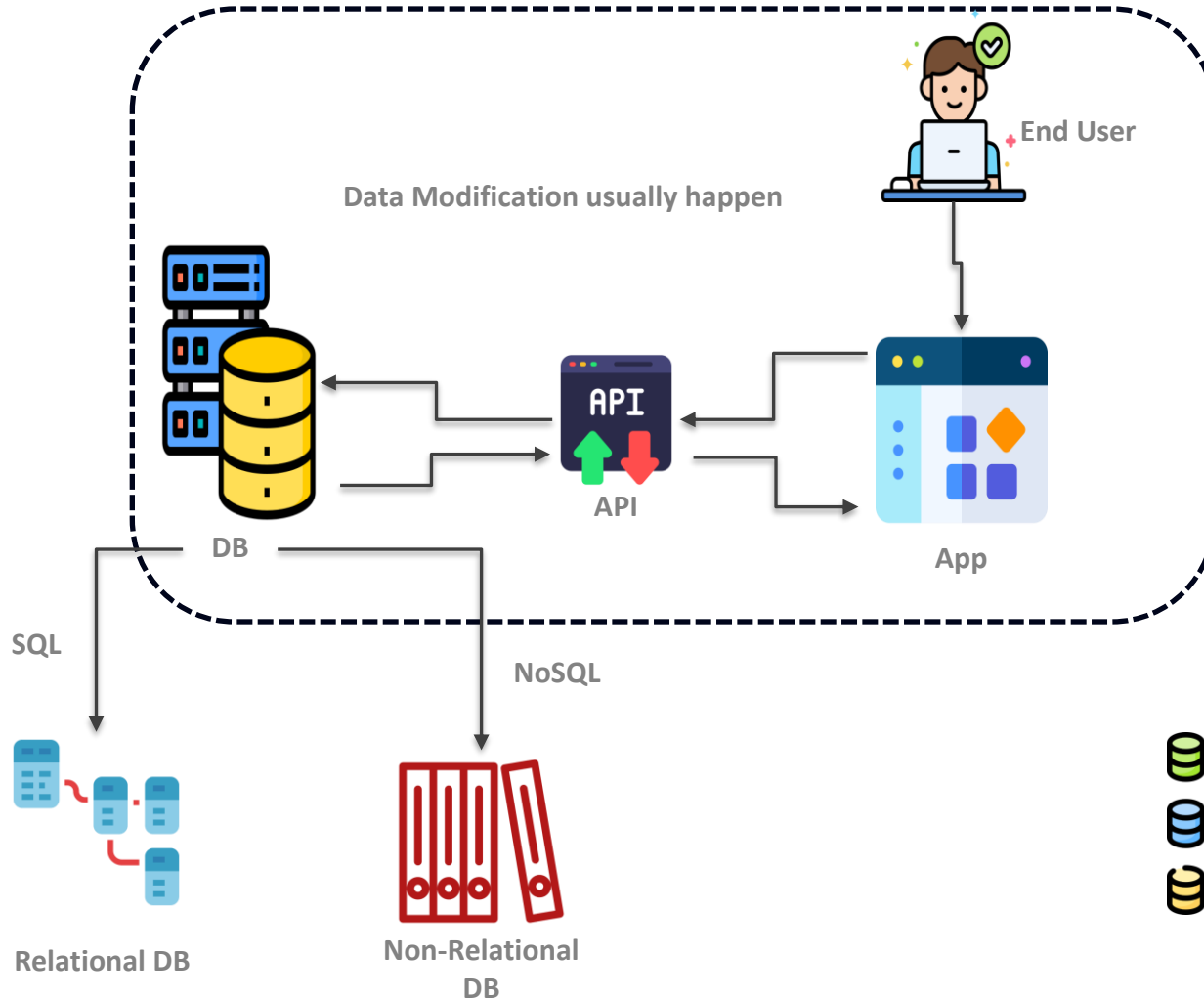
Part Three: Data Storage and Management

Course Road Map

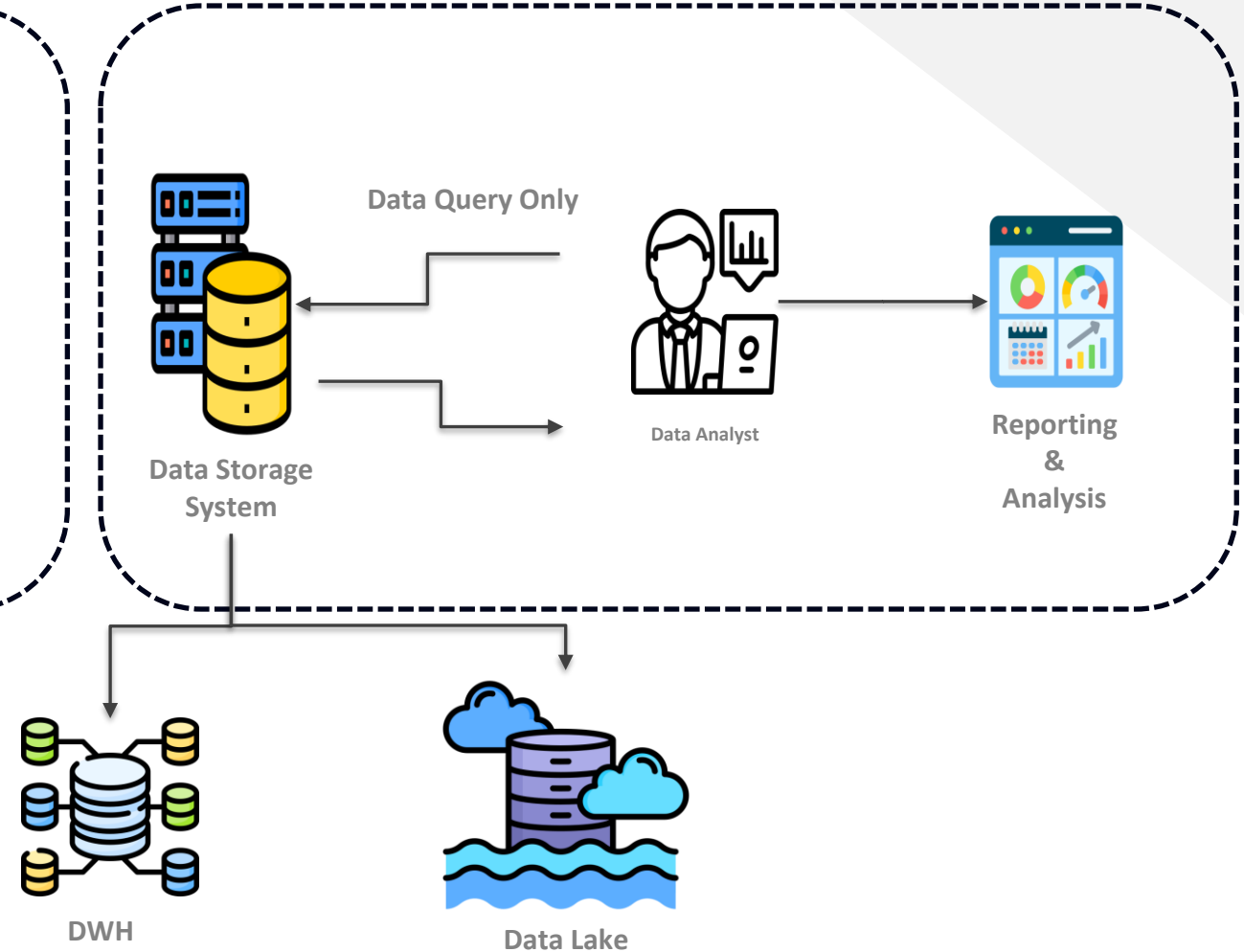


OLTP vs OLAP

OLTP (Online Transaction Processing)



OLAP (Online Analytical Processing)



Relational databases (SQL)

- Definition
- Characteristics of SQL Databases
- SQL Database Advantages
- Common SQL Systems



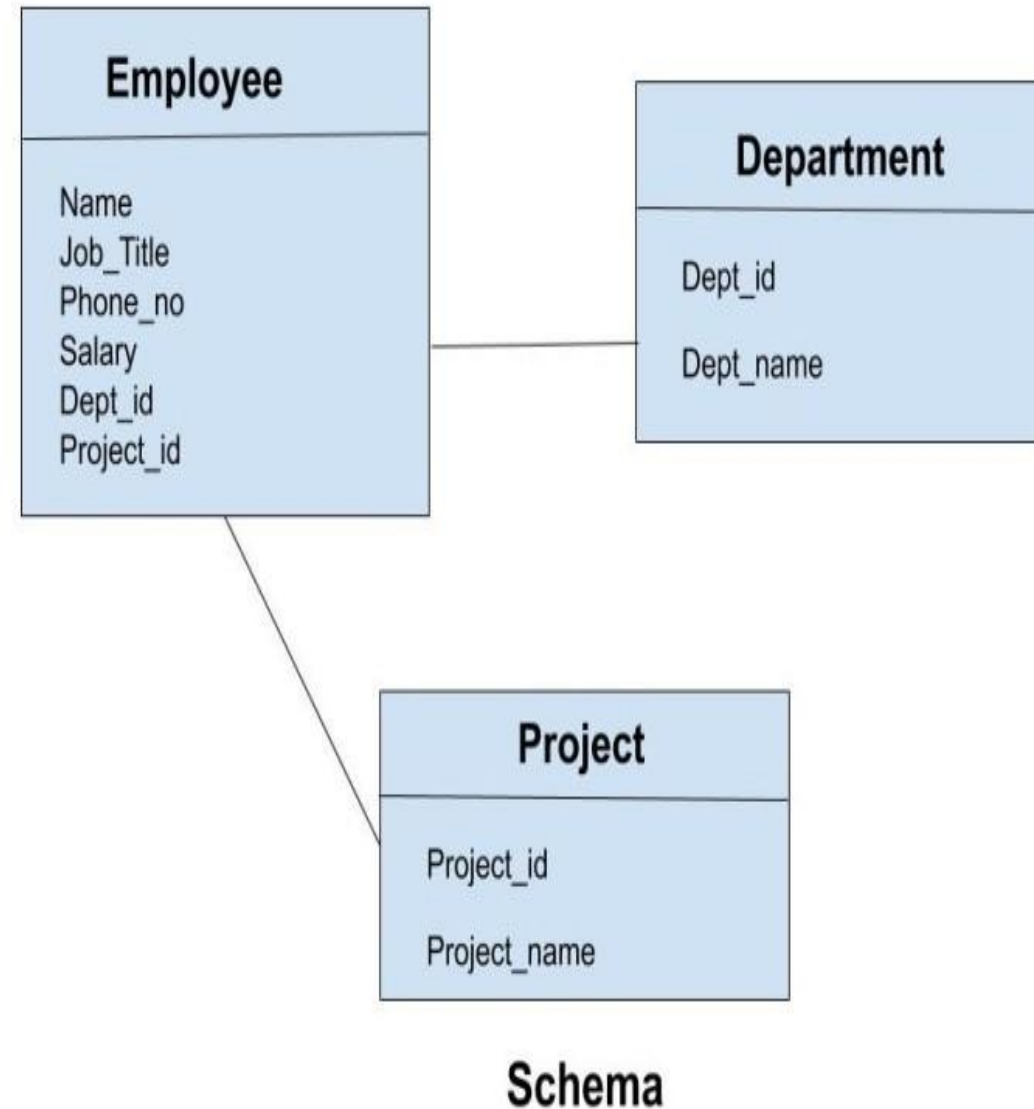
SQL

- Structured Query Language, is a standard programming language specifically designed for managing and manipulating relational databases.
- It is widely used for querying, updating, and managing data stored in relational database management systems (RDBMS).



SQL Characteristics

- Schema:
defines the structure of the database, including tables, their fields, data types, available values, and relationships.
- Tables:
The foundational units of a database, representing entities (e.g., information about customers, products, or transactions), with rows containing individual records for each entity (e.g., a specific customer) and columns storing specific attributes about the entity (e.g., the customer's name or email).



SQL Advantages

Property	Description
Atomicity	Ensures that each attempted set of changes to a database is treated as a single, indivisible unit of work. Either the whole unit is committed to the database, or none of them are if any part of the transaction fails.
Consistency	Ensures that a transaction ends with the database in a valid state. The database must satisfy a set of integrity constraints both before and after the transaction.

SQL Advantages

Property	Description
Isolation	Ensures that simultaneous transactions do not lead to inconsistencies in the database. Each transaction appears to execute in isolation, unaware of other transactions running. Isolation prevents interference between transactions and maintains their integrity.
Durability	Durability guarantees that once a transaction is committed, its effects persist even in the event of a system failure. The changes made by the transaction are permanently stored in the database, and they survive system crashes or power outages.

Non-Relational Databases (NoSQL)

- Definition
- NoSQL Database Types
- NoSQL Database Advantages



NoSQL

- NoSQL, or "Not Only SQL" databases are designed to handle large volumes of data, high user loads, and flexible data models, making them suitable for modern applications that require scalability and agility.

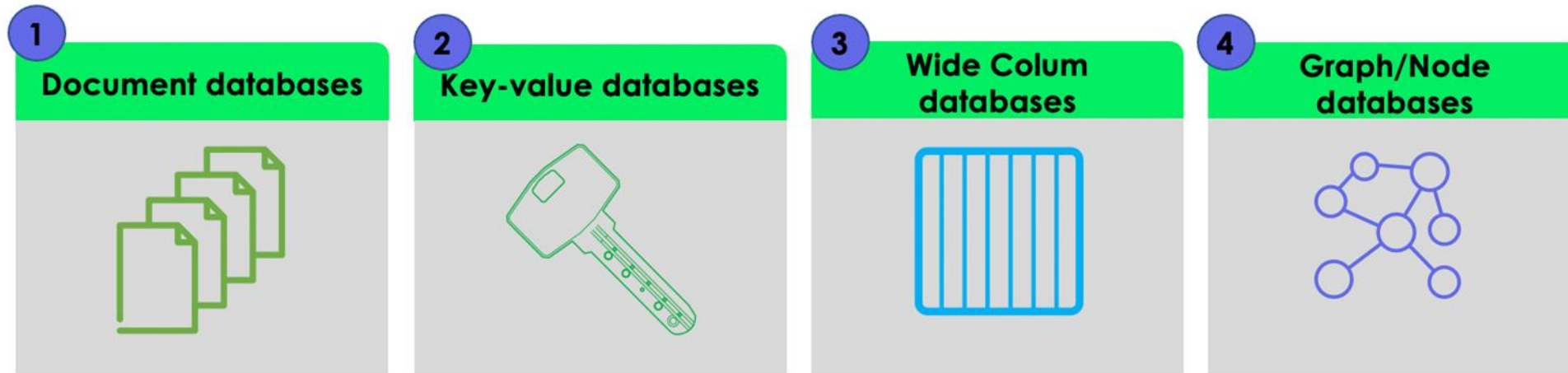


Question

- Imagine you're building a social media app where users can upload images, post messages, and send friend requests.
Would SQL or NoSQL be better - and why?



NoSQL Database Types



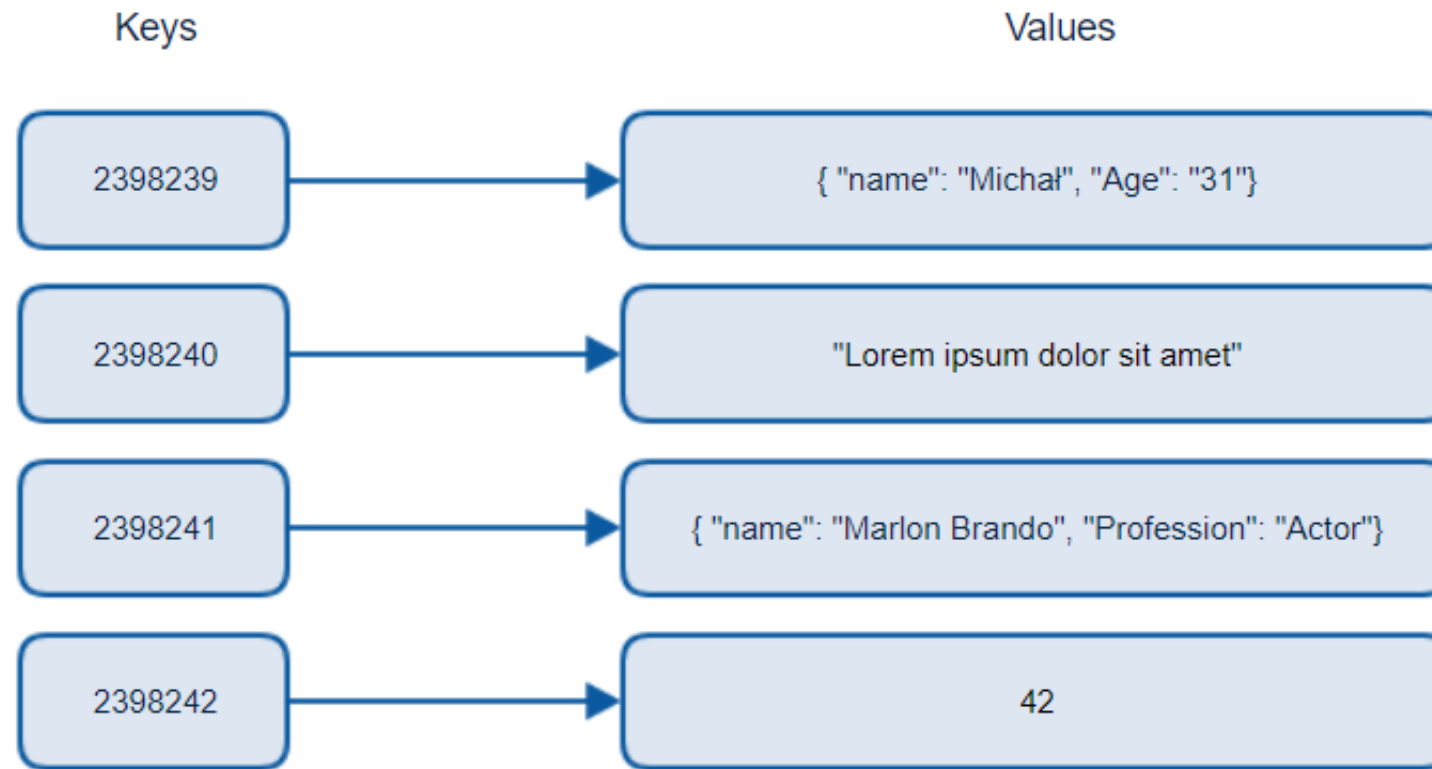
Document Database

```
{
  "_id": "tomjohnson",
  "firstName": "Tom",
  "middleName": "William",
  "lastName": "Johnson",
  "email": "tom.johnson@digitalocean.com",
  "department": ["Finance", "Accounting"],
  "socialMediaAccounts": [
    {
      "type": "facebook",
      "username": "tomjohnson"
    },
    {
      "type": "twitter",
      "username": "@tomjohnson"
    }
  ]
}
```

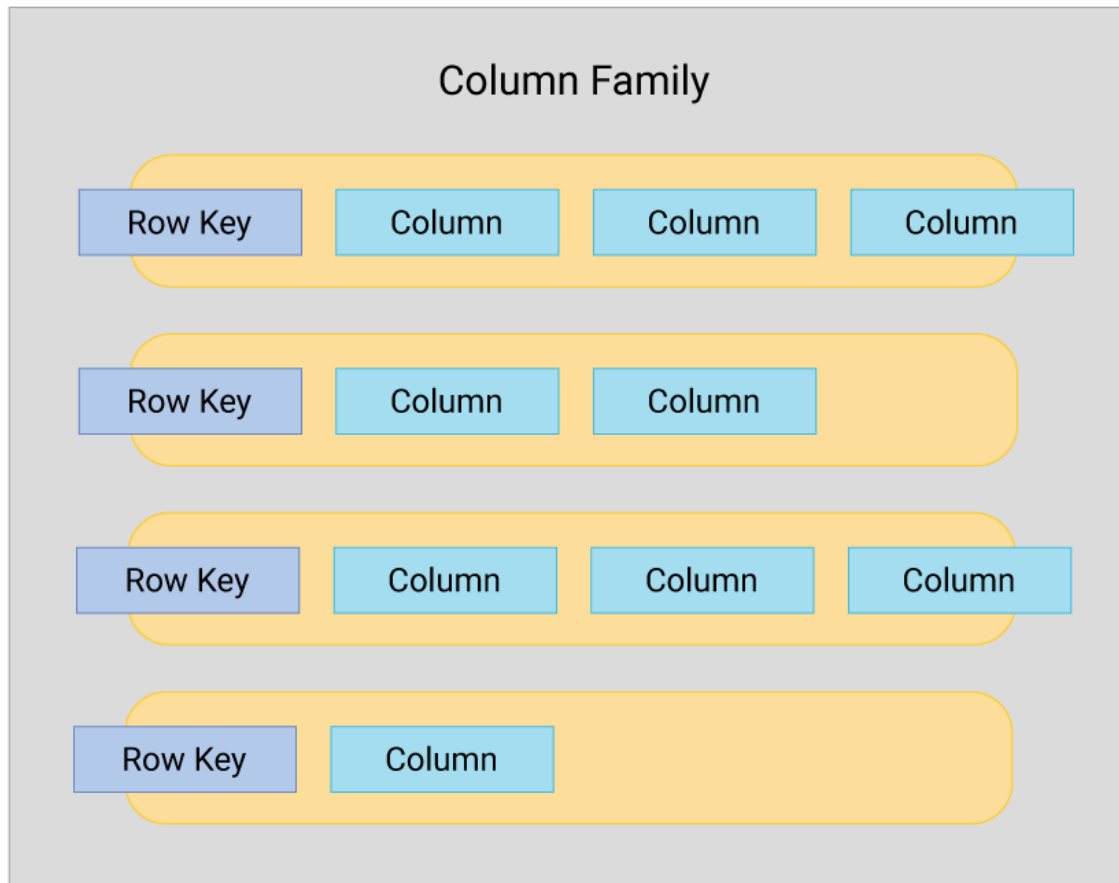
```
{
  "_id": "sammyshark",
  "firstName": "Sammy",
  "lastName": "Shark",
  "email": "sammy.shark@digitalocean.com",
  "department": "Finance"
}
```

```
{
  "_id": "tomjohnson",
  "firstName": "Tom",
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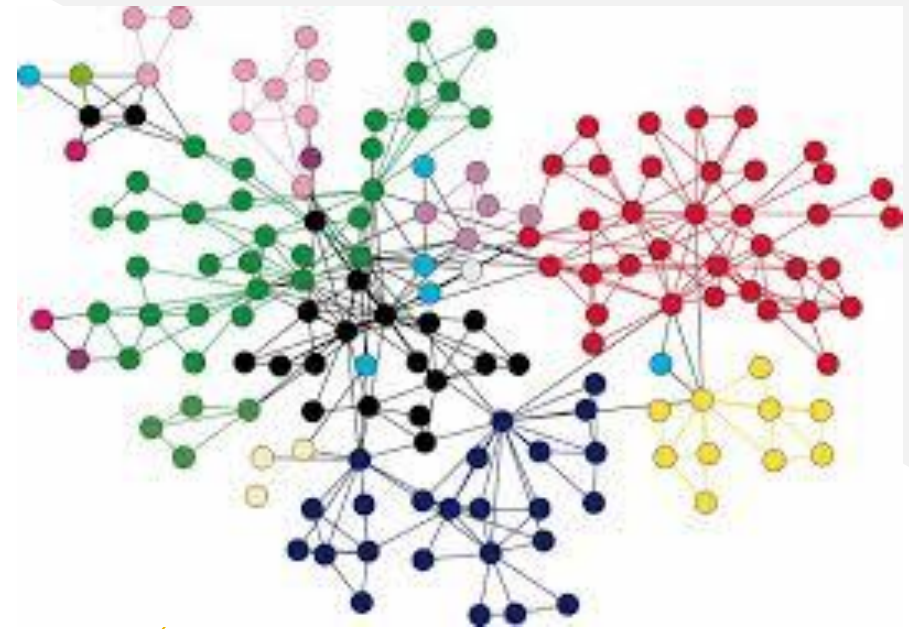
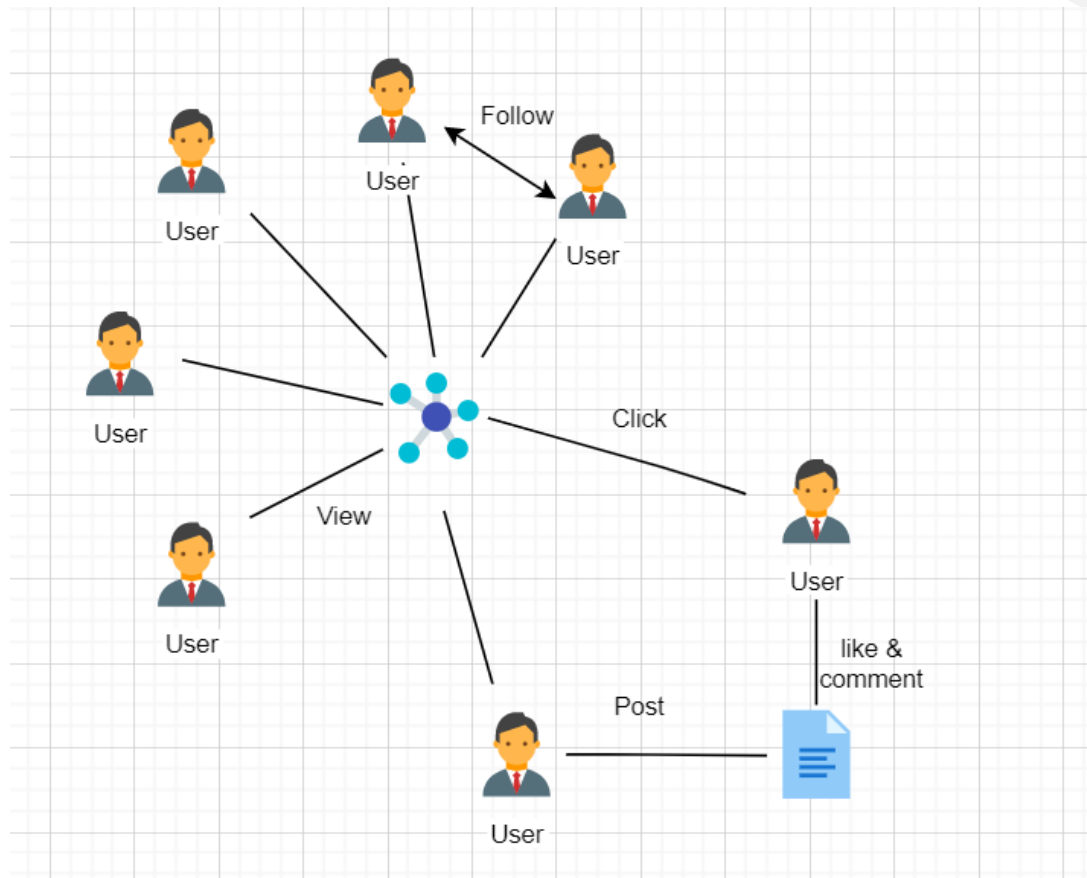
Key Value Database



Wide Column Database



Graph Database



NoSQL Database Types

Database Type	Description	Examples
Document Stores	Store data as flexible, JSON-like documents. Suitable for complex, hierarchical data with dynamic schemas.	MongoDB
Key-Value Database	Store data as simple key-value pairs. Efficient for quick data access and fast response times.	Redis, Amazon DynamoDB
Graph Database	Focus on relationships between data entities, ideal for scenarios like social networks or recommendation systems.	Neo4j, Amazon Neptune
Column-Family Database	Store data in columns rather than rows, beneficial for large-scale distributed systems and applications with frequent read/write operations.	Apache Cassandra, HBase

Question

- Let's say you're building a music app - you want to store user preferences, fast search history, and social connections.
Which types of NoSQL databases would you consider using - and why?



Use Cases

Industry	Use Case
Finance	Managing transactional data and customer records with strict consistency (ACID properties).
Retail	Handling complex relationships related to products, sales, customers, and suppliers.
Government and Public Sector	Managing citizen records and public services while adhering to regulatory requirements.

Industry	Use Case
Social Media	Handling large volumes of unstructured data such as user profiles, posts, and interactions.
Logistics and Supply Chain	Real-time tracking of shipments, managing dynamic data sources across the supply chain.
Gaming	Managing player data, leaderboards, and real-time analytics for online multiplayer games.

Data Warehouses And Data Lakes

- Data Warehouse Definition
- Data Warehouse Design And Concepts
- Data Warehouse VS Data Lakes

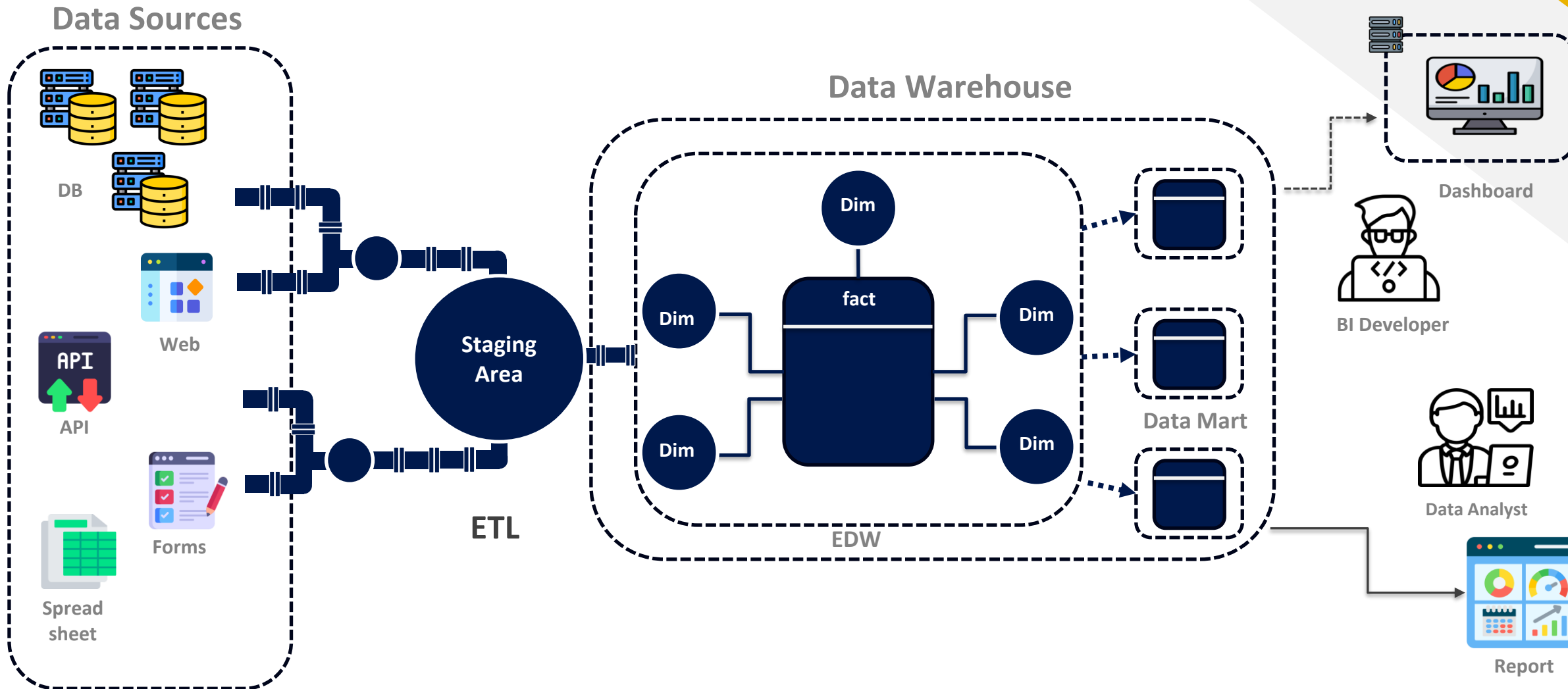


Data Warehouse Definition

- Data Warehouse is a centralized repository where an organization can store substantial amounts of data from multiple source systems and locations
- A heterogeneous collection of different data sources organized under a unified schema.



Data Warehouse

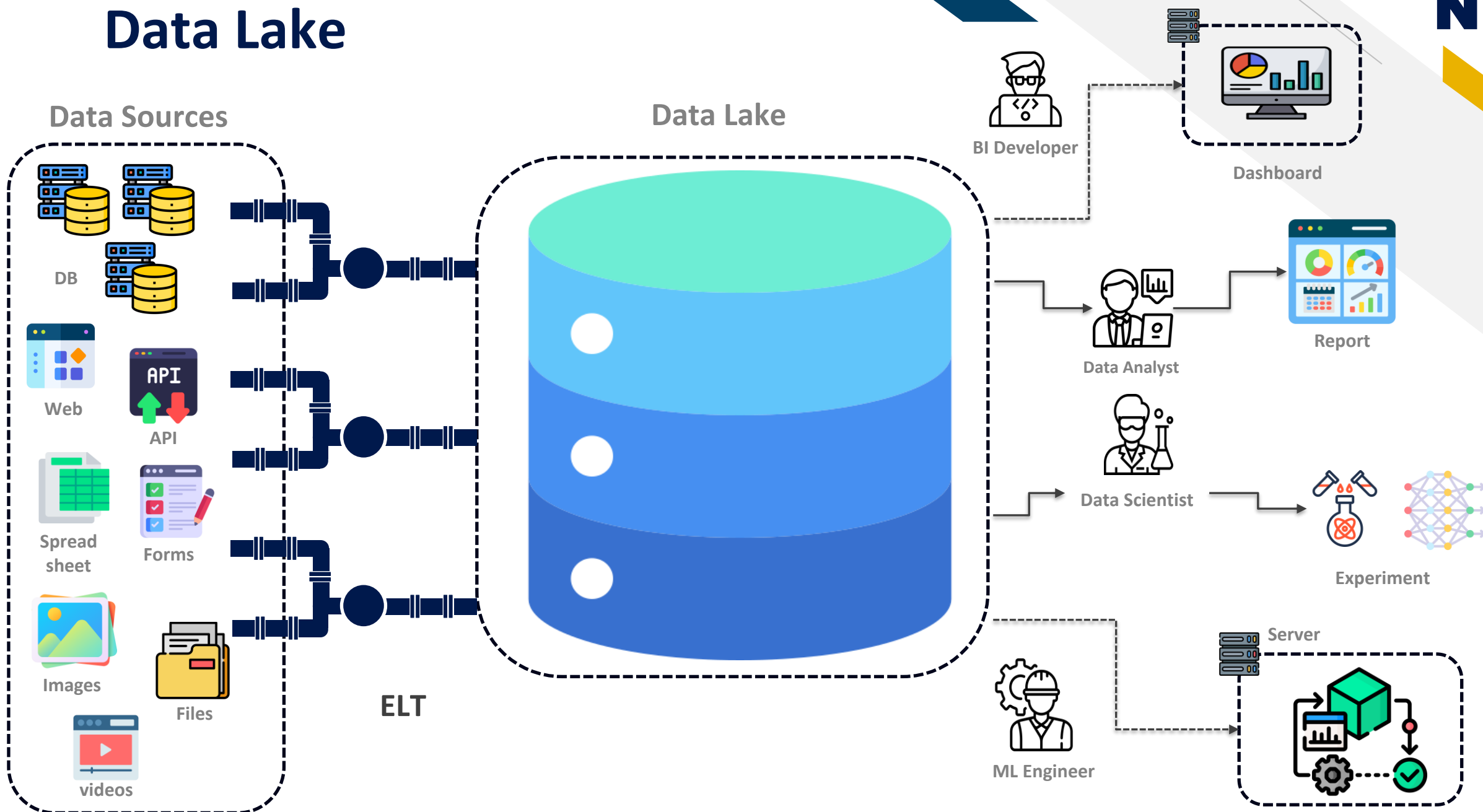


Data Lakes

- A data lake is a repository that enables you to store all the structured and unstructured data you collected at any scale.
- Data lakes allow you to run different types of analytics, from visualizations and dashboards to big data processing, real-time analytics, and machine learning that can help you make better decisions.

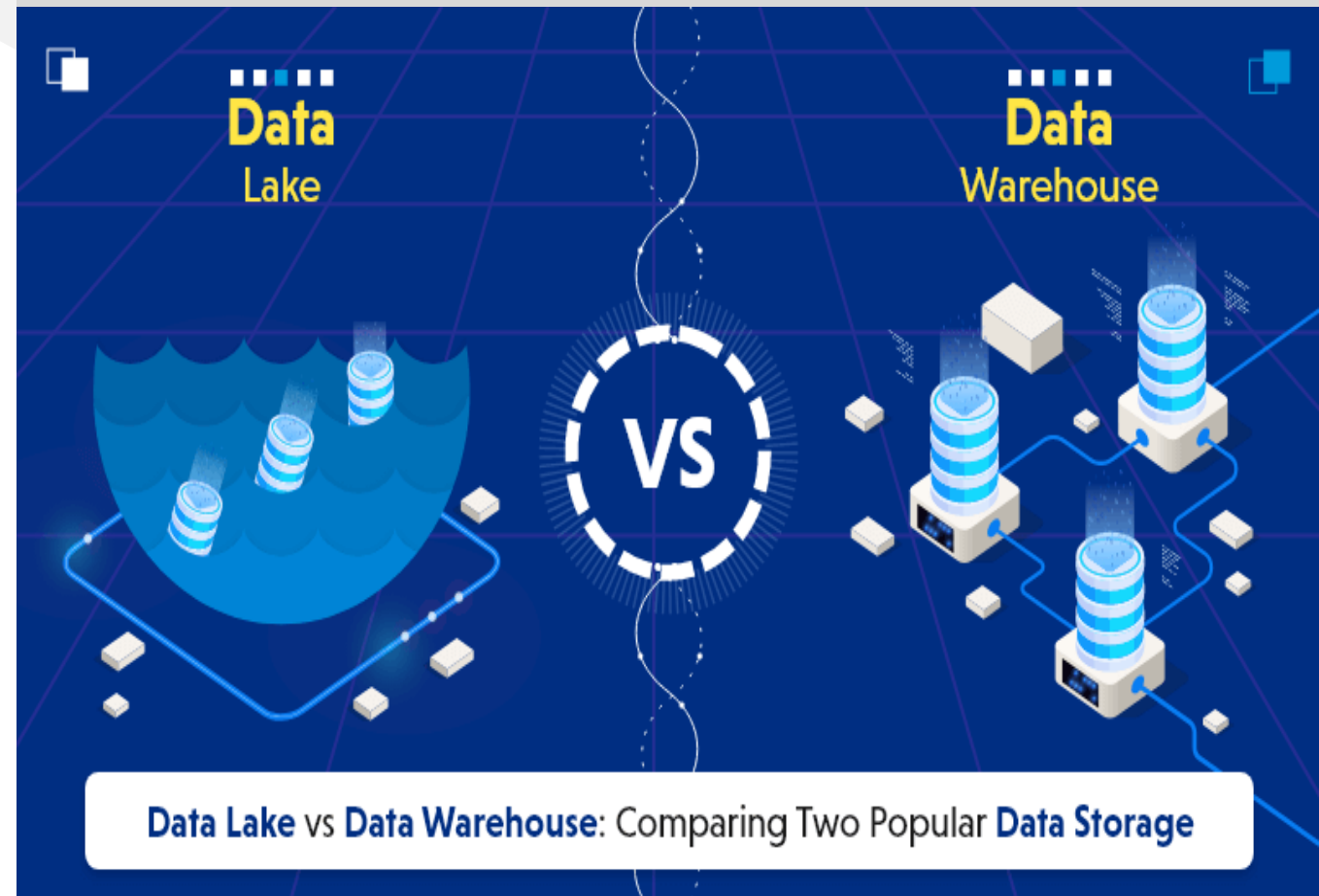


Data Lake



Data Warehouses VS Data Lakes

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DATA LAKE

vs

DATA WAREHOUSE

Data



unstructured

Users



Data Scientists,
Data Analysts

Use cases



Stream Processing,
Machine Learning,
Real time analysis

Raw

Data Lakes contain unstructured, semi structured and structured data with minimal processing. It can be used to contain unconventional data such as log and sensor data

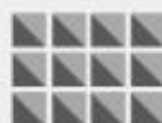
Large

Data Lakes contain vast amounts of data in the order of petabytes. Since the data can be in any form or size, large amounts of unstructured data can be stored indefinitely and can be transformed when in use only

Undefined

Data in data lakes can be used for a wide variety of applications, such as Machine Learning, Streaming analytics, and AI

Data



Structured

Users



Business Analysts

Use cases



Batch Processing,
BI, Reporting

Refined

Data Warehouses contain highly structured data that is cleaned, pre-processed and refined. This data is stored for very specific use cases such as BI.

Smaller

Data Warehouses contain less data in the order of terabytes. In order to maintain data cleanliness and health of the warehouse, Data must be processed before ingestion and periodic purging of data is necessary

Relational

Data Warehouses contain historic and relational data, such as transaction systems, operations etc



APIs & Web Scraping

- APIs Definition
- APIs Examples: Twitter API, Google Maps API.
- How to connect to an API and retrieve data.
- Python Application (Web Scraping)
- Ethical and legal considerations.



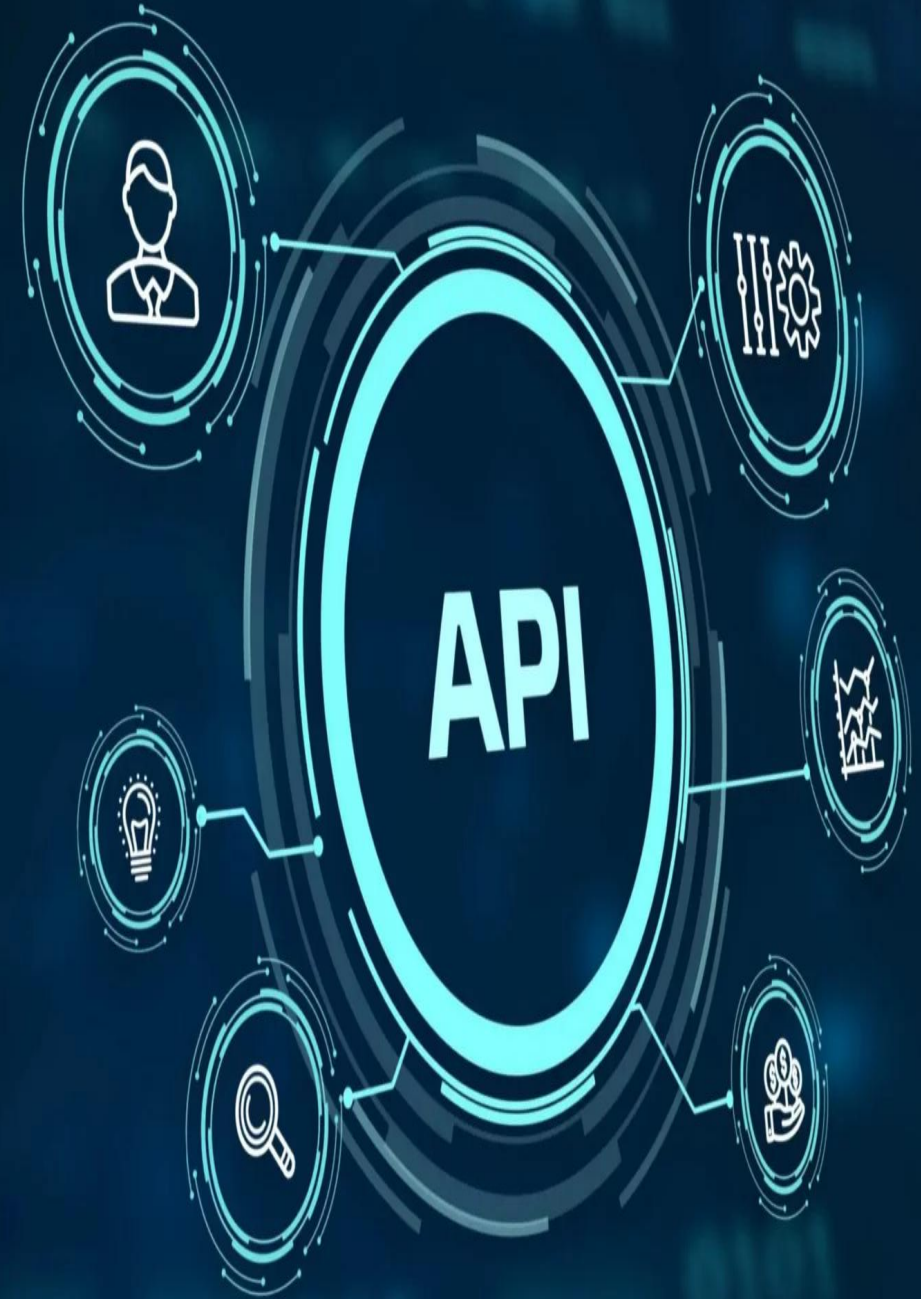
APIs

- An API, or Application Programming Interface, is a set of rules and protocols for building and interacting with software applications.
- It defines the methods and data formats that applications can use to communicate with each other.

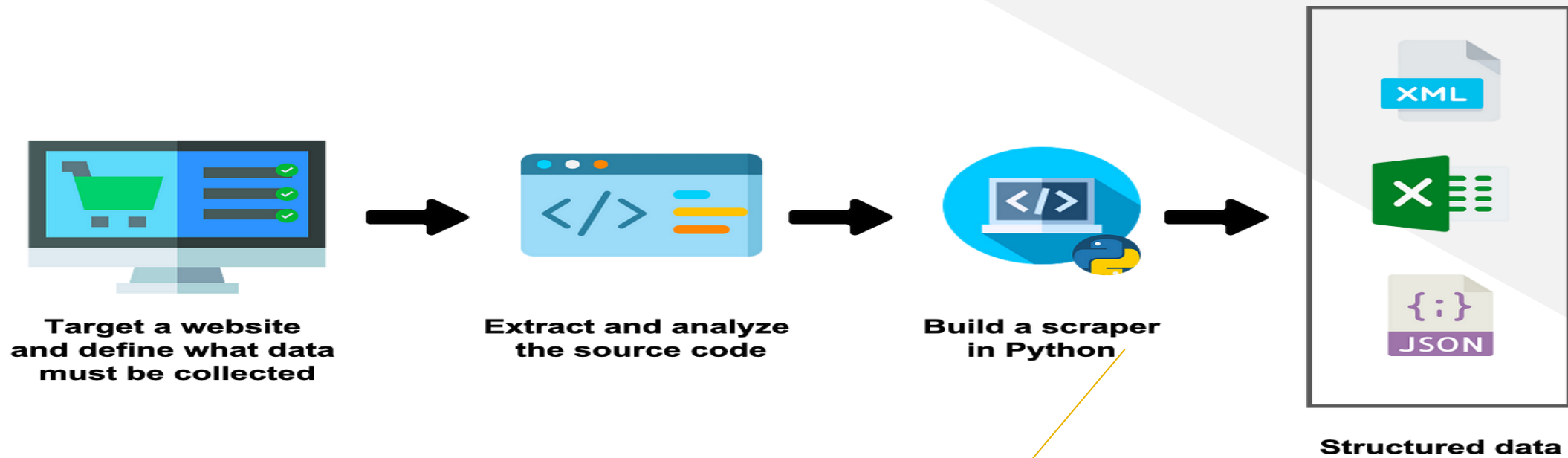


Types of APIs

- Web APIs:
 - REST (Representational State Transfer)
 - SOAP (Simple Object Access Protocol)
- Library-Based APIs:
 - Sets of routines, protocols, and tools for building software and applications.
 - For example, the standard libraries in languages like Java, C++, or Python.



(Web Scraping with Python)



Questions





Thank You.



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